

TOLBO, Mongolia

WMO: 442170

Lat: 48.417N

Lon: 90.300E

Elev: 2101

StdP: 78.50

Time Zone: 7.00 (E07)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.7	-32.4	-37.5	0.1	-34.3	-35.4	0.2	-32.0	8.2	-11.4	6.9	-13.1	0.2	190	0.730

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	22.7	11.8	21.2	10.9	19.7	10.3	13.4	20.0	12.3	18.7	11.4	17.7	2.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.8	10.4	16.1	9.5	9.5	15.0	8.4	8.9	14.2	44.4	19.9	41.4	18.7	38.6	17.8	18.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	6.9	6.0	-36.1	25.1	3.3	1.5	-38.5	26.2	-40.4	27.1	-42.2	27.9	-44.6	29.0		
(5)			-36.3	14.8	3.2	2.4	-38.6	16.5	-40.4	17.9	-42.2	19.3	-44.6	21.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.3	-20.9	-17.3	-8.1	0.1	5.3	11.7	13.5	11.7	5.6	-1.8	-10.7	-17.6	(6)
(7)		DBStd	12.68	5.92	5.88	5.31	4.55	4.08	3.10	2.62	2.71	3.68	4.55	5.29	5.62	(7)
(8)		HDD10.0	4750	958	763	562	298	152	18	3	15	136	366	622	857	(8)
(9)		HDD18.3	7534	1216	997	821	548	406	198	150	207	381	624	872	1115	(9)
(10)		CDD10.0	259	0	0	0	0	6	70	113	67	4	0	0	0	(10)
(11)		CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0	(11)
(12)		CDH23.3	25	0	0	0	0	0	5	16	4	0	0	0	0	(12)
(13)		CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0	(13)
(14)	Wind	WSAvg	2.0	1.1	1.5	2.0	2.5	2.9	2.3	2.1	2.1	2.1	2.0	1.6	1.3	(14)
(15)	Precipitation	PrecAvg	166	3	2	5	9	14	29	42	30	12	8	7	5	(15)
(16)		PrecMax	239	9	4	15	18	28	80	108	65	24	16	13	9	(16)
(17)		PrecMin	111	1	1	0	4	4	8	12	8	4	1	2	1	(17)
(18)		PrecStd	34	2	1	4	4	7	15	22	14	6	4	3	2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-3.2	-1.8	6.9	14.2	19.6	24.1	25.1	23.8	20.2	12.6	5.5	-3.3	(19)
(20)			MCWB	-5.4	-4.8	0.7	5.4	8.6	12.4	12.8	11.3	8.7	4.7	0.6	-6.1	(20)
(21)		2%	DB	-7.4	-4.1	4.6	11.9	17.0	21.7	23.1	21.8	17.4	10.0	2.7	-5.6	(21)
(22)			MCWB	-9.2	-6.8	-1.0	3.9	8.0	11.3	12.3	10.6	7.9	3.2	-0.9	-7.5	(22)
(23)		5%	DB	-9.2	-5.9	2.7	9.8	15.0	20.0	21.6	20.2	15.4	7.8	0.0	-7.2	(23)
(24)			MCWB	-10.7	-8.1	-2.0	2.8	6.9	10.5	11.6	9.9	7.2	2.2	-3.0	-8.9	(24)
(25)		10%	DB	-11.4	-7.7	0.9	7.8	13.1	18.4	20.0	18.7	13.5	5.5	-2.4	-8.9	(25)
(26)			MCWB	-12.7	-9.7	-3.0	2.0	5.6	9.9	11.0	9.6	6.3	0.8	-5.0	-10.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.0	-4.6	1.1	6.8	10.4	14.1	15.3	13.7	10.7	5.5	0.9	-5.6	(27)
(28)			MCDWB	-3.6	-2.5	5.2	12.8	18.1	21.7	22.3	20.4	17.2	10.7	4.6	-3.6	(28)
(29)		2%	WB	-9.0	-6.6	-0.4	5.0	8.6	12.4	13.8	12.2	9.1	4.0	-0.8	-7.4	(29)
(30)			MCDWB	-7.5	-4.4	3.9	9.9	15.2	19.4	20.5	18.8	15.5	9.3	2.6	-5.5	(30)
(31)		5%	WB	-10.8	-8.0	-1.7	3.5	7.4	11.5	12.9	11.2	7.9	2.6	-2.8	-8.9	(31)
(32)			MCDWB	-9.2	-5.9	2.3	8.7	14.0	18.0	19.1	17.8	14.2	7.3	0.0	-7.2	(32)
(33)		10%	WB	-12.7	-9.7	-2.9	2.3	6.3	10.6	11.9	10.3	6.6	1.1	-4.8	-10.4	(33)
(34)			MCDWB	-11.4	-7.8	0.8	7.5	12.5	16.8	18.0	16.8	12.8	5.3	-2.4	-8.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.5	12.7	13.2	11.3	10.2	10.4	11.8	12.0	10.7	10.7	10.5	(35)	
(36)			MCDBR	12.0	14.0	14.5	14.5	13.5	13.2	13.5	14.9	15.9	14.5	13.4	11.1	(36)
(37)			MCWBR	10.5	11.9	10.2	8.5	7.1	5.4	5.3	6.2	8.2	8.8	10.0	9.6	(37)
(38)		5% WB	MCDBR	11.7	13.6	12.7	12.4	12.2	11.4	11.7	12.3	14.5	13.5	12.8	10.8	(38)
(39)			MCWBR	10.3	11.6	9.1	8.0	7.0	5.2	5.5	6.1	8.3	8.6	9.7	9.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.222	0.249	0.294	0.340	0.334	0.342	0.336	0.317	0.283	0.266	0.242	0.224	(40)	
(41)		taud	2.337	2.261	2.183	2.194	2.288	2.357	2.398	2.426	2.481	2.446	2.377	2.319	(41)	
(42)		Ebn at Noon	882	920	920	907	925	918	918	924	931	894	843	826	(42)	
(43)		Edh at Noon	67	92	120	133	126	119	112	104	88	76	64	60	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.71	2.80	4.27	5.49	6.41	6.60	6.42	5.74	4.66	2.94	1.76	1.33	(44)	
(45)		RadStd	0.07	0.10	0.21	0.24	0.41	0.36	0.37	0.22	0.22	0.21	0.12	0.08	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (6 neighbors)		N/A	N/A	N/A	N/A	-0.63	-1.00	N/A	N/A	N/A

Nomenclature: See separate page